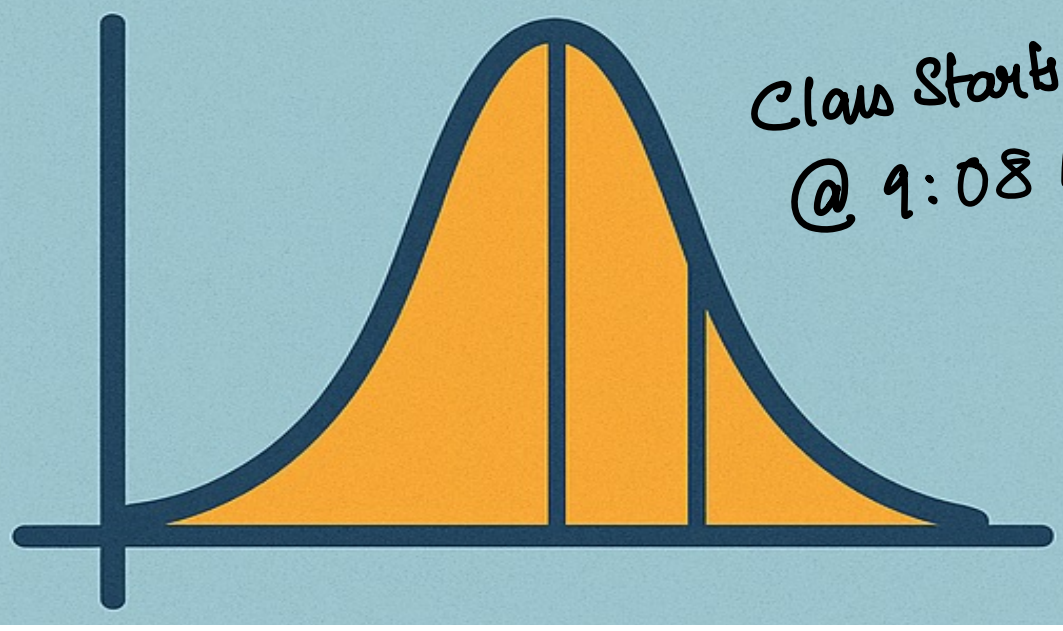


PROBABILITY AND STATISTICS





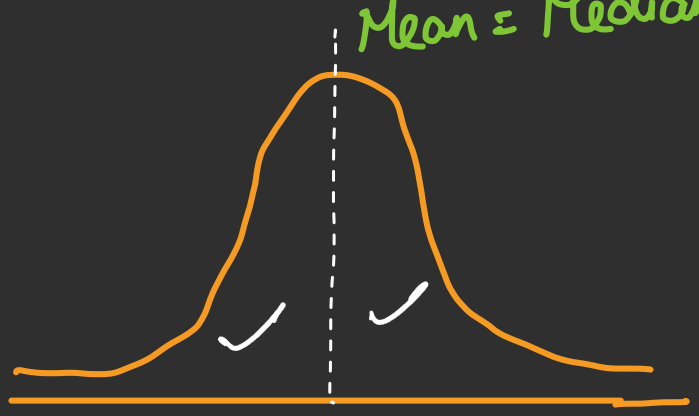
CLT and CI

#Agenda

1. Central limit theorem → Theory
2. Application of CLT [Code]
3. Confidence Interval ✓
4. Normal Distribution
   - Standard Normal Distribution

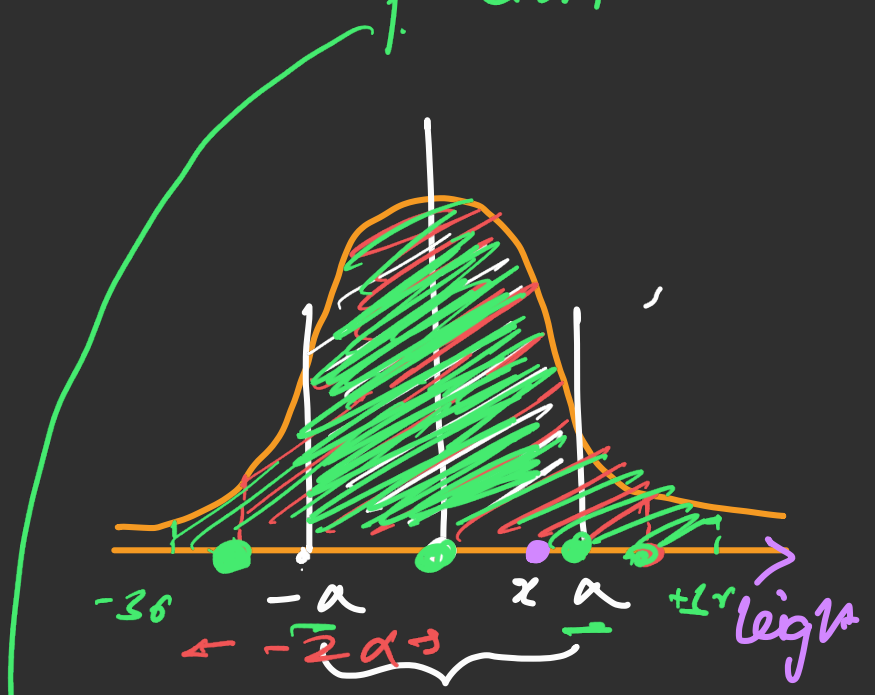
Recap

Normal dist / Gaussian



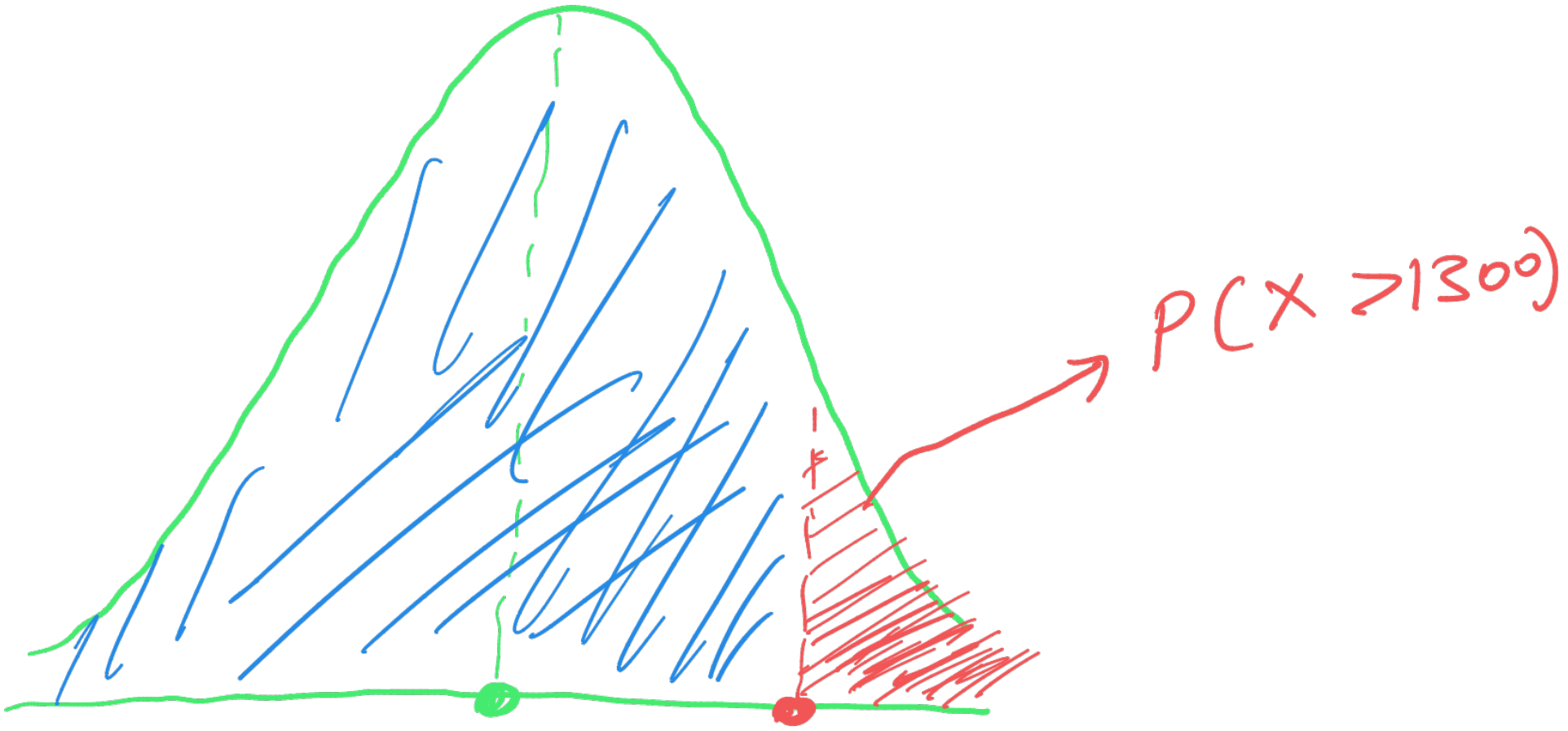
Emp Rule: 68 – 95 – 99 Rule

$$Z = \frac{x - \mu}{\sigma}$$



z ⟶ value on x axis

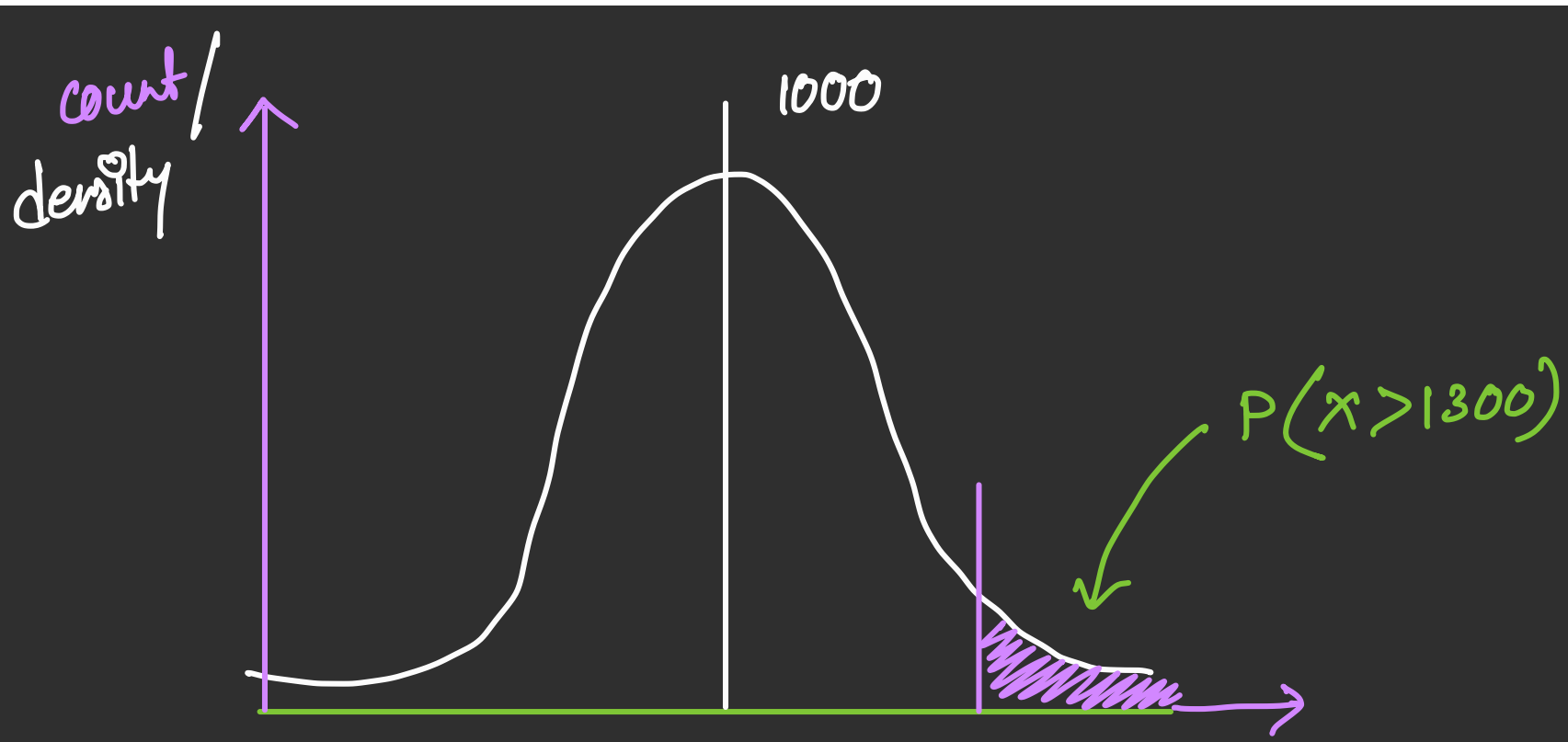
ppf ⟶ opp of z score / cdf



$P(X \leq 1300)$ → norm. cdf

We always calculate cdf from left to right

$$P(X > 1300) = 1 - \text{norm. cdf}(X \leq 1300)$$



$P(x \geq 1300)$

$1 - P(x \leq 1300)$

$z = \frac{1300 - 1000}{200} \rightarrow 1.5$

norm.cdf(z)

1 - norm.cdf (1.5)

$P(173 < X < 300)$ = 1) pdf (Calculates the prob within a certain range)

norm.pdf( )

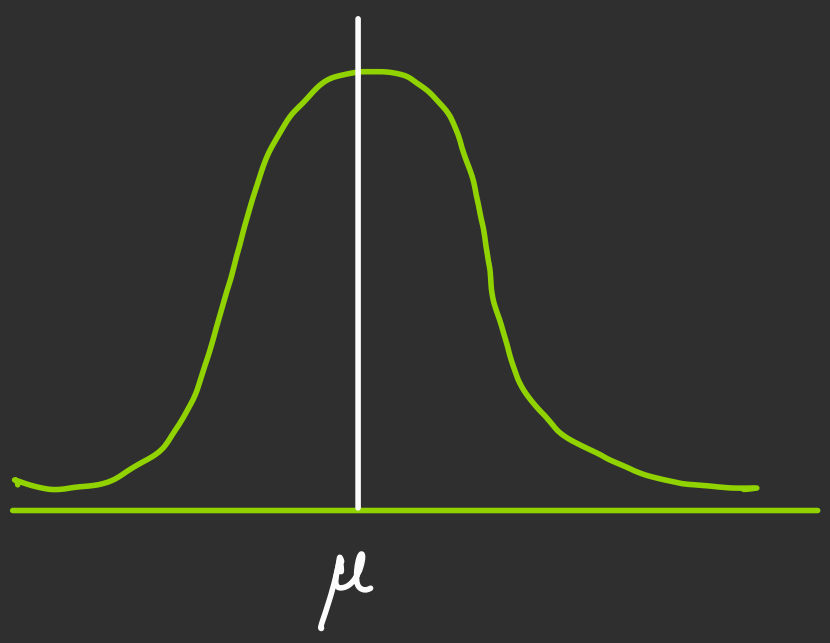
$P(X < 300)$ = 2) cdf → (Area Under the curve upto that point)

norm.cdf(z)

$P(X > 1200)$ = 3) 1 − cdf → (Cdf calculates from L → R)

1 − norm.cdf(z)

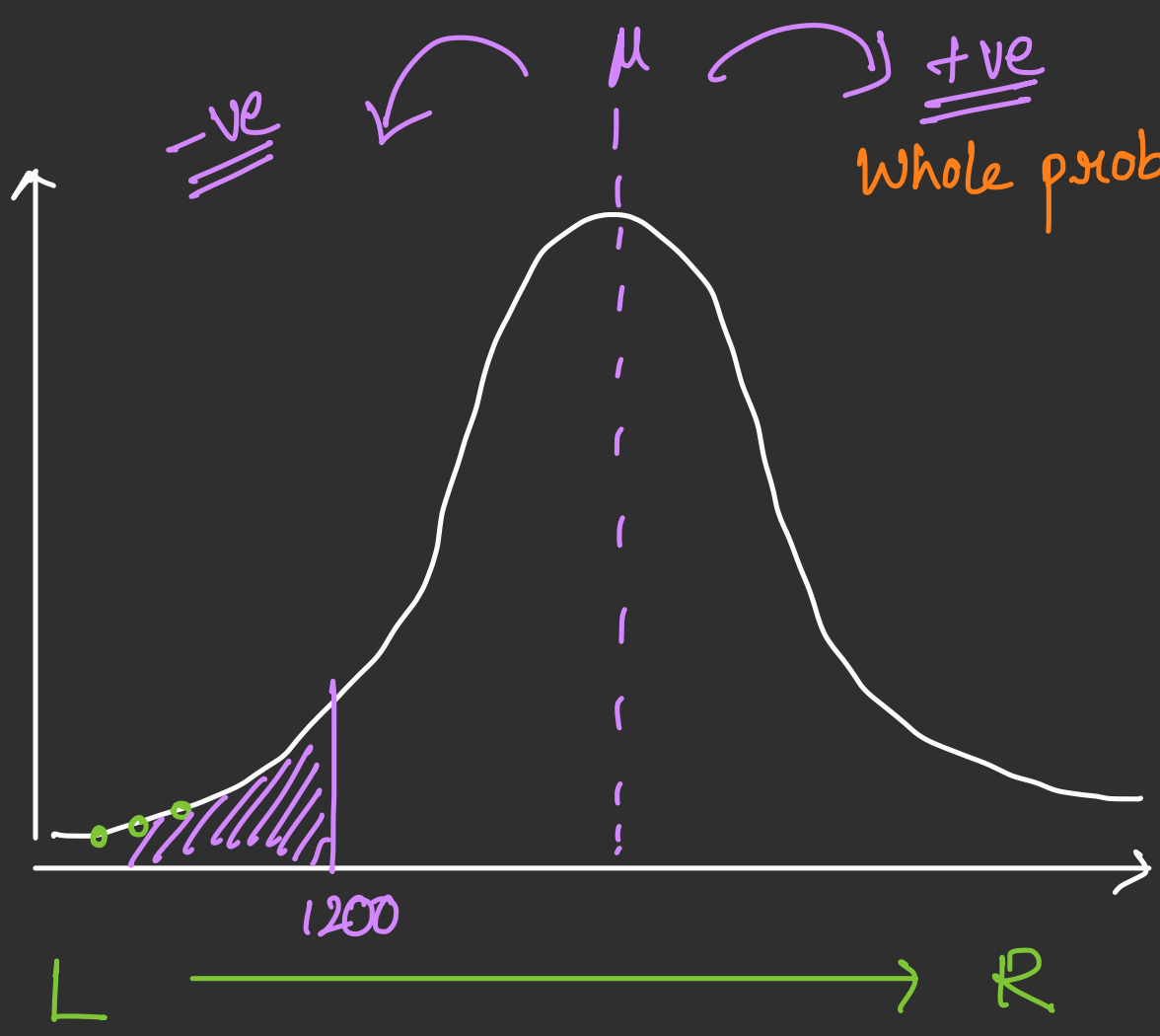
Standard Normal



→ $\mu = 0$, $\sigma = 1$ — Standard Normal

Parameter of a Normal dist → μ, σ

$z = \frac{x - \mu}{\sigma}$ } → $\mu = 0, \sigma = 1$ } Standard Normal graph



1 − norm.cdf(z)

P(x < 1200)

↳ norm.cdf(z)

Central limit Theorem

Q) What is the avg weight of the JB Basket?

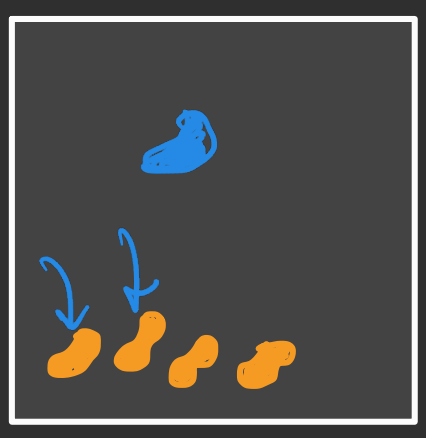
pop mean = ?

$S_1 = [w_1, w_2, w_3 \cdots, w_{20}] = \bar{x}_1$ — $\bar{x}_1$

$S_2 = [w_1, w_2, w_3 \cdots w_{20}] = \bar{x}_2$

$S_3 = [w_1, w_2, w_3 \cdots w_{20}] = \bar{x}_3$

$S_{1000} = [w_1, w_2, w_3 \cdots w_{20}] = \bar{x}_{1000}$



Pop siz = ?

10B

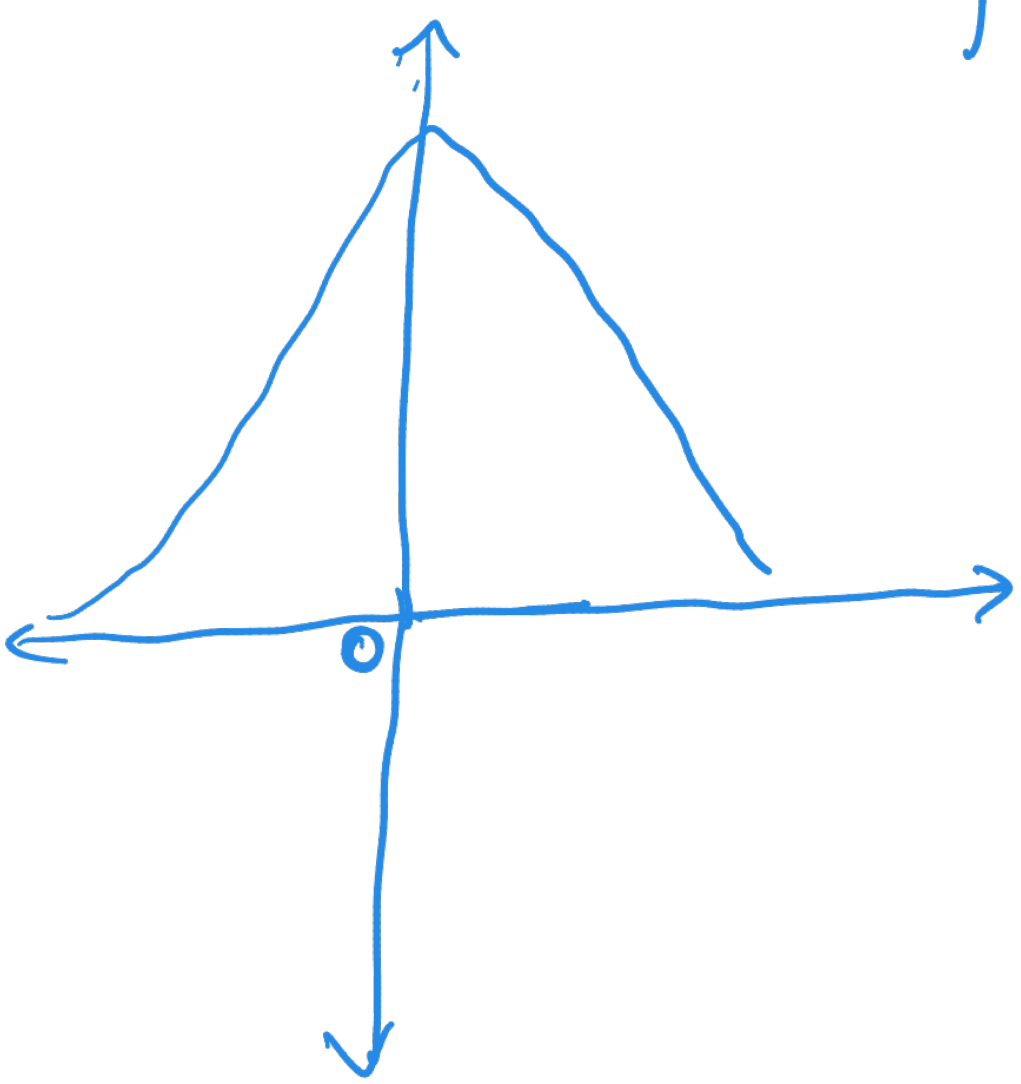
"sampling dist"

mean

$$\left[\frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \cdots \bar{x}_{1000}}{1000}\right]$$

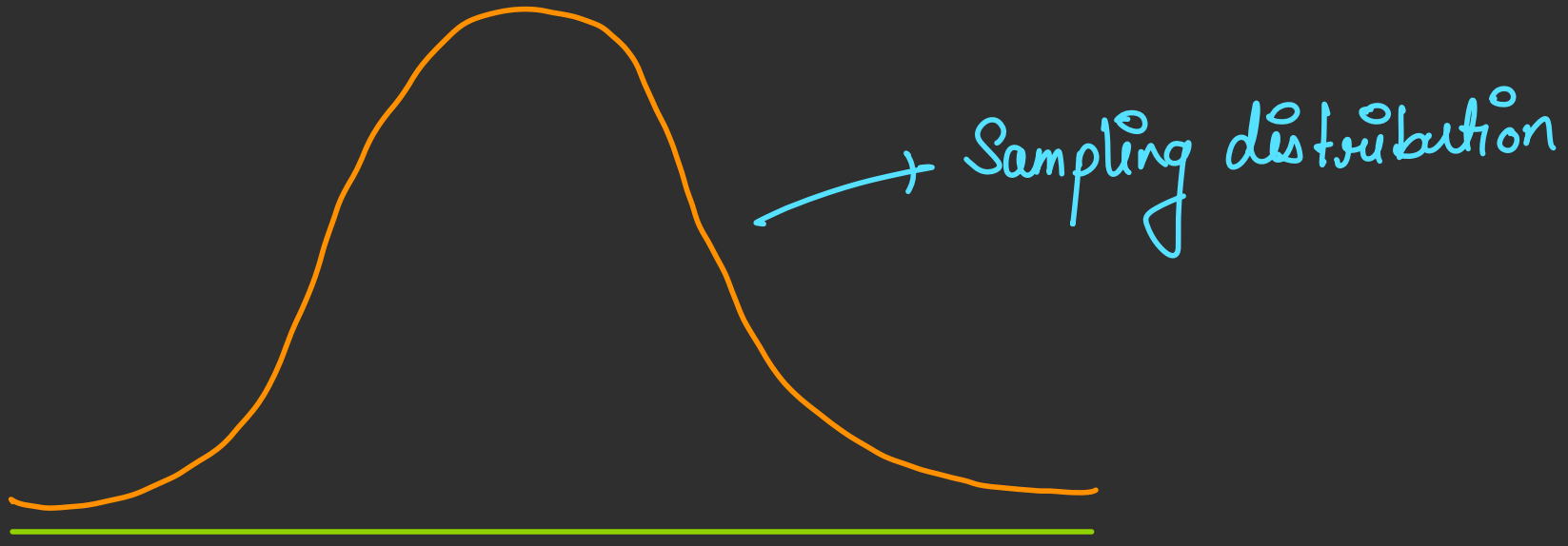
SE ↓

The center of the bell curve is at zero
by symmetrical around zero



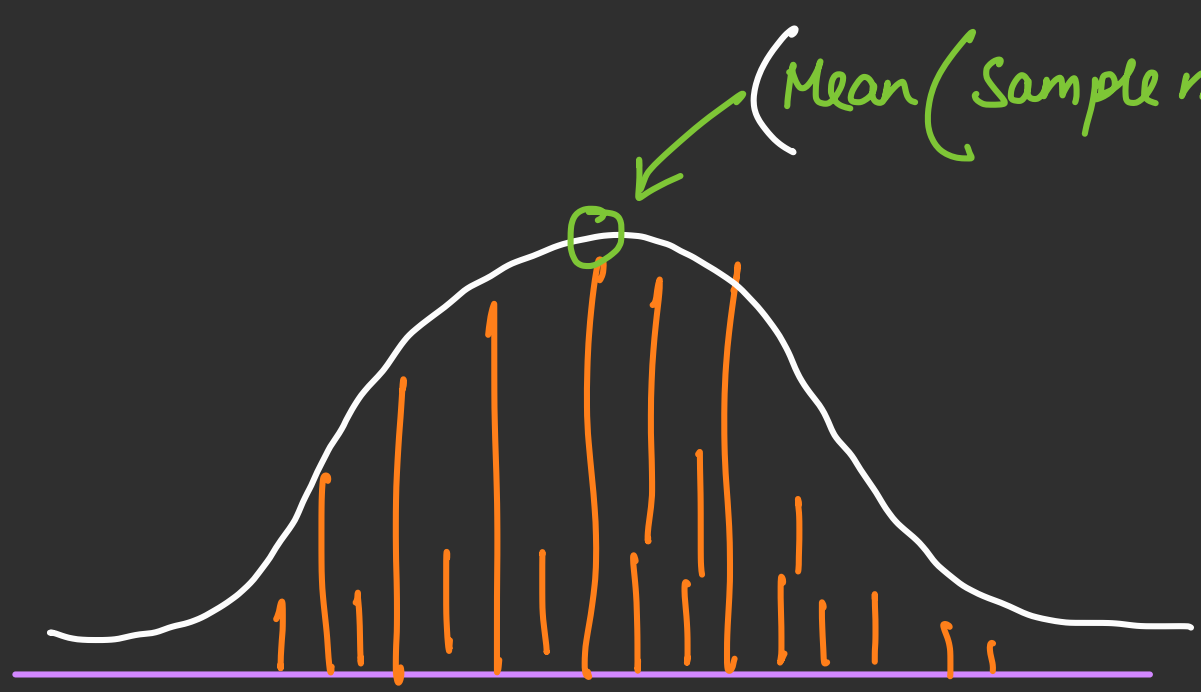
collection of sample means

$$[\bar{x}_1, \bar{x}_2, \bar{x}_3 \ldots, \bar{x}_{100}]$$



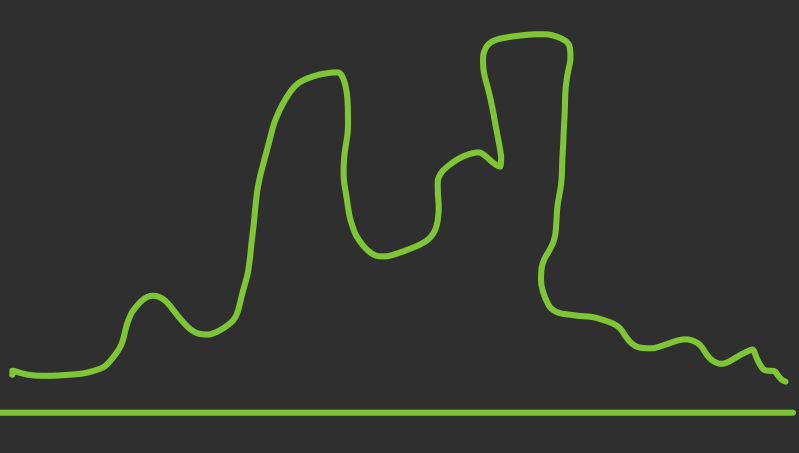
Means of the sample mean

(Mean (Sample mean)) ≈ pop^{n} mean

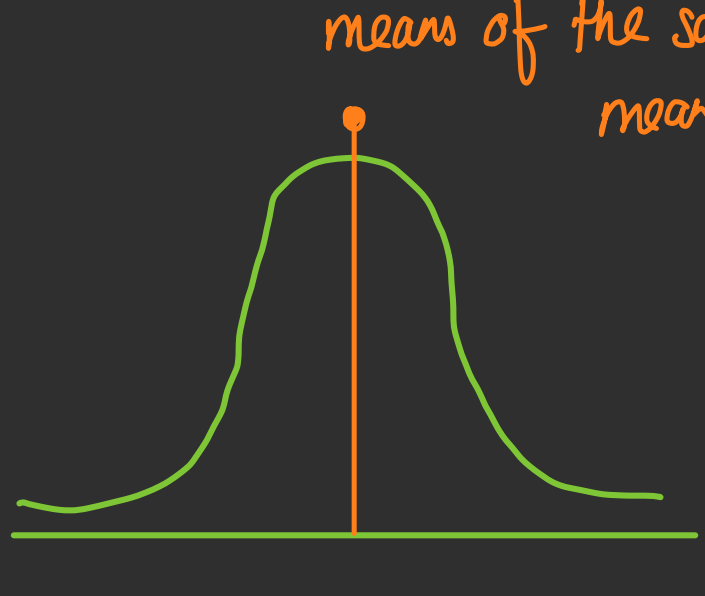


$\bar{x}_1, \bar{x}_2, \bar{x}_3 \ldots \bar{x}_{1000}$

1000



CLT



"Not Normal"

pop^{n} mean = sample means

(means of the sample means)

Sample size ⇒ 4

Sample size = 40

Sample size = 80

↓ Standard Error (SE) = $\frac{\sigma}{\sqrt{n}}$ ↑

40 80

↳ "Law of large Number"

$$SE \propto \frac{1}{\sqrt{n}}$$

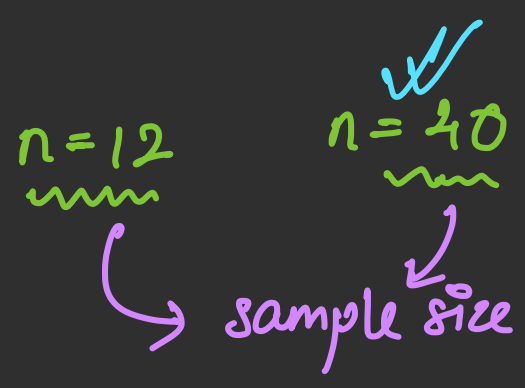
$n \uparrow$ $SE \downarrow$ ✓

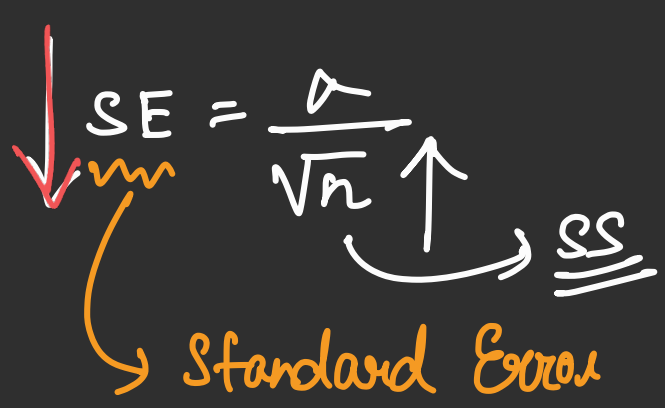
$n \downarrow$ $SE \uparrow$

Sample Size

Q) What is the right sample size to take?







Law of Large No

(n > as large as possible)

[n ≥ 30] → sufficiently large SS

Application property of CLT

The population mean is approx equal to the means of the sample means "given that" sample size is large enough and your pop^n mean & std are finite

$n > 30$

$\bar{x}$

#Conditions

To apply the central limit theorem, the following conditions must be met:

Randomization:

1. Data should be randomly sampled, ensuring every population member has an equal chance of being included.

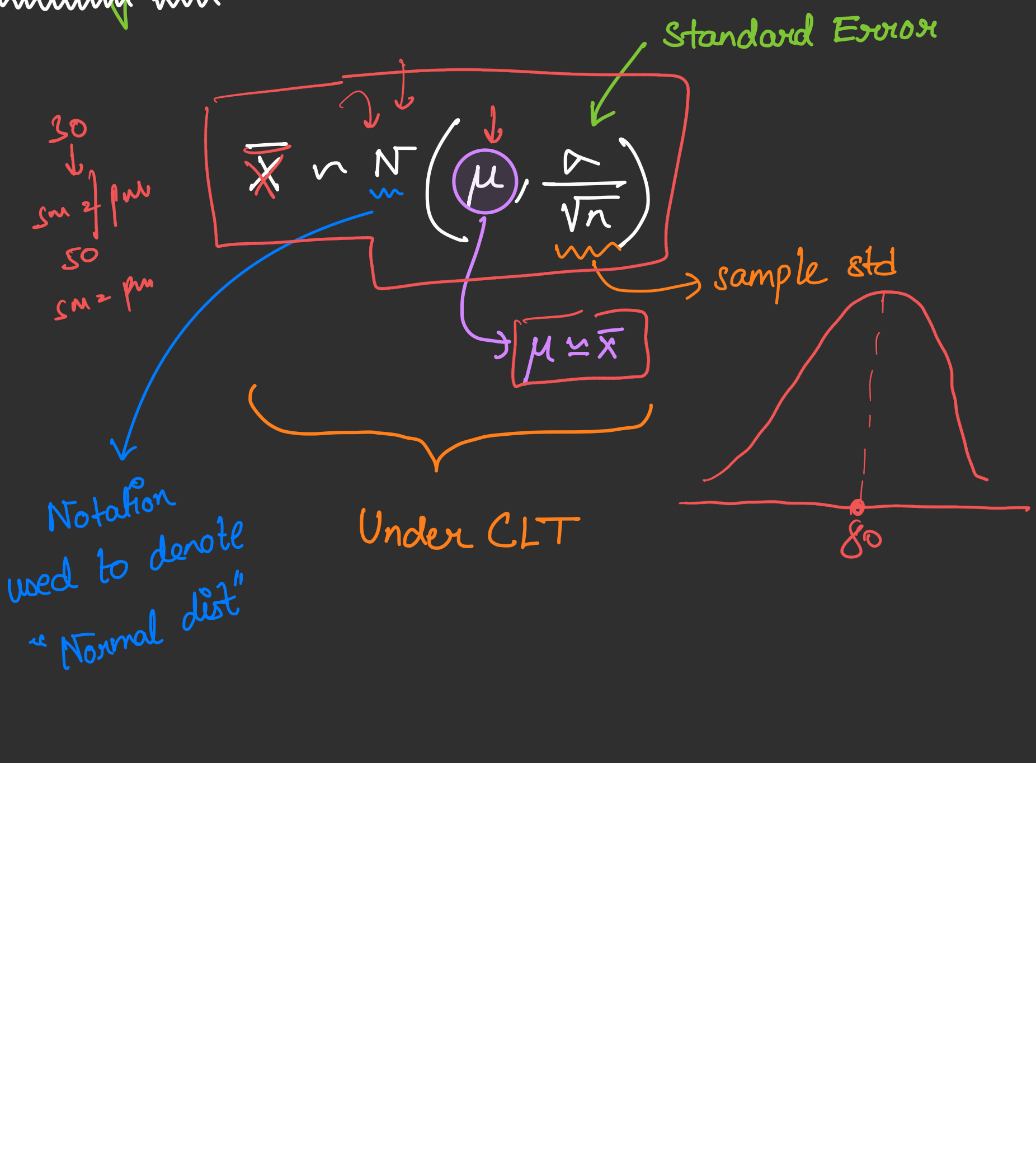
Independence:

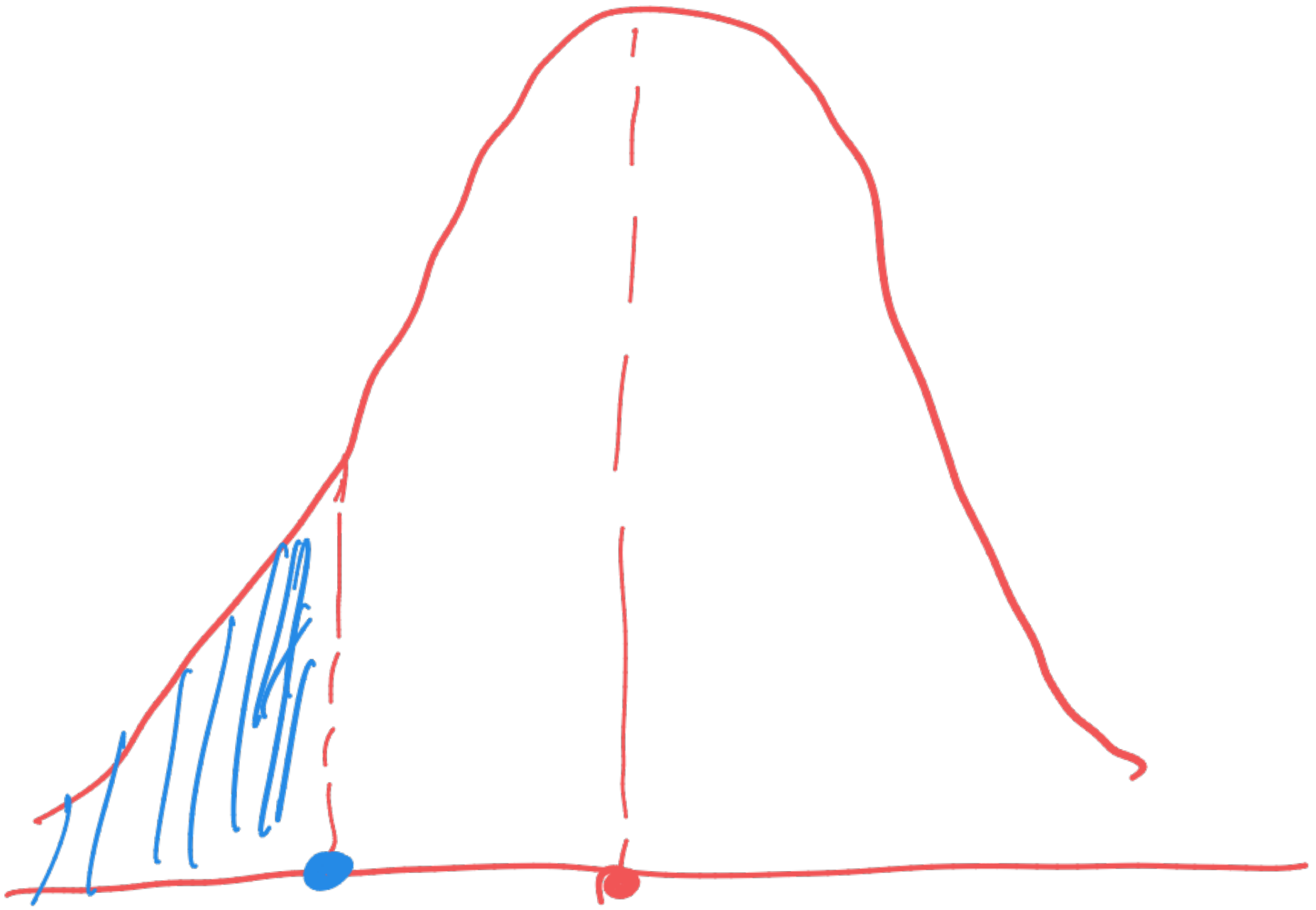
1. Each sample value should be independent, with one event's occurrence not affecting another.
2. Commonly met in probability sampling methods, which independently select observations.

Large Sample Condition:

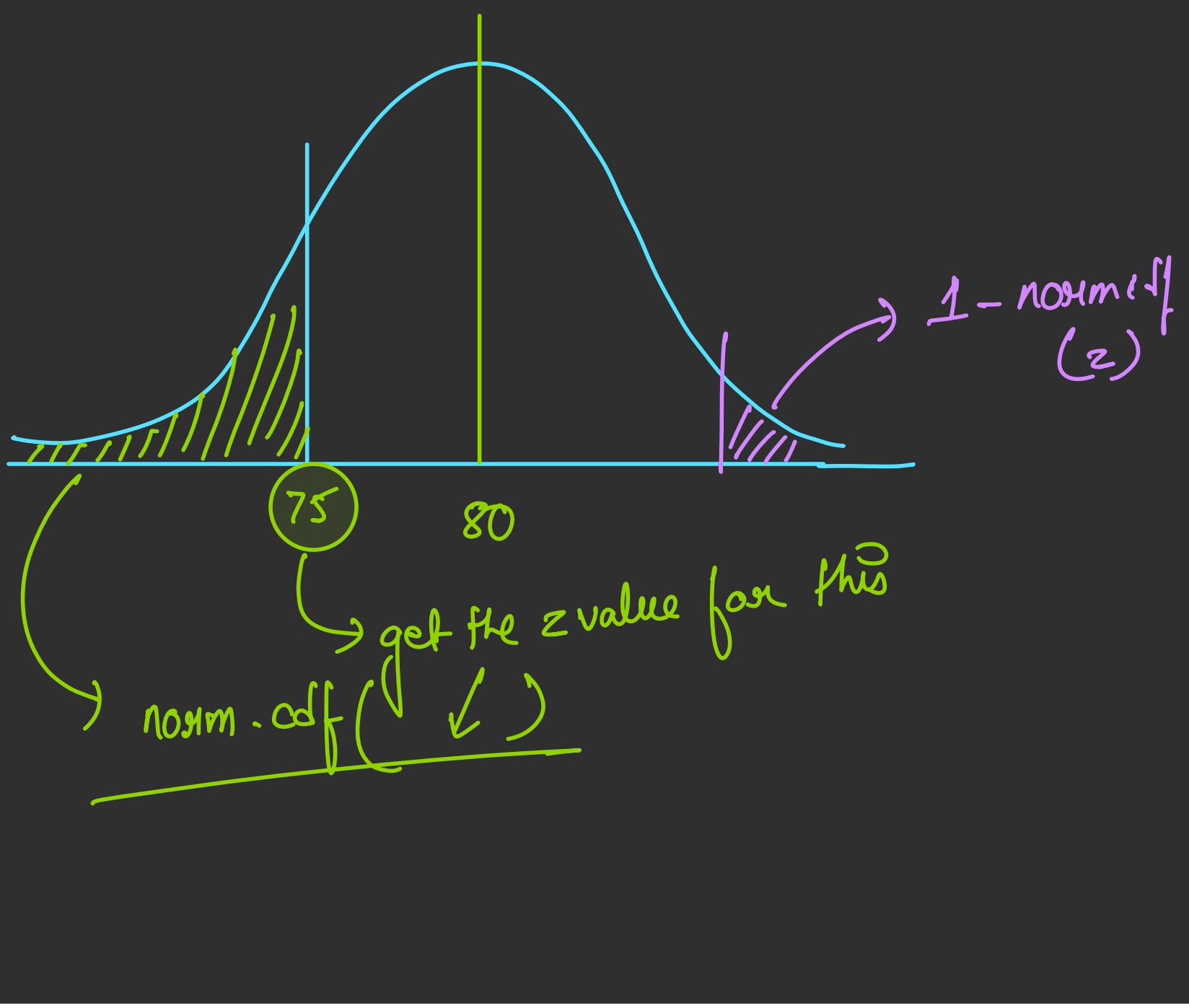
1. A sample size of 30 or more is generally considered "sufficiently large."
2. This threshold can vary slightly based on the population distribution's shape.

Summary CLT

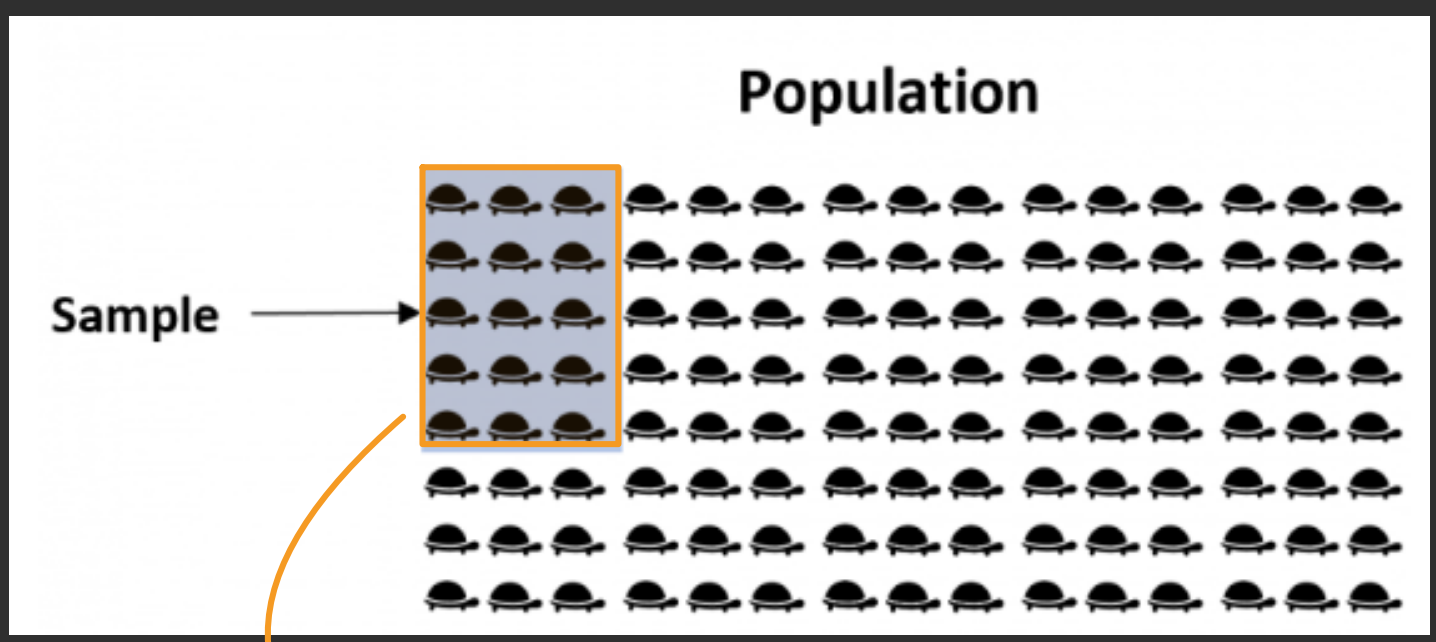

Standard Error
30
sm ≠ pm
50
sm = pm
$\bar{x} \sim N\left(\mu, \frac{\Delta}{\sqrt{n}}\right)$
sample std
$\mu \simeq \bar{x}$
Under CLT
Notation
used to denote
"Normal dist"
80



P(x ≤ 75) → calc normal calc


1 - norm.cdf (z)
75
80
get the z value for this
norm.cdf( )

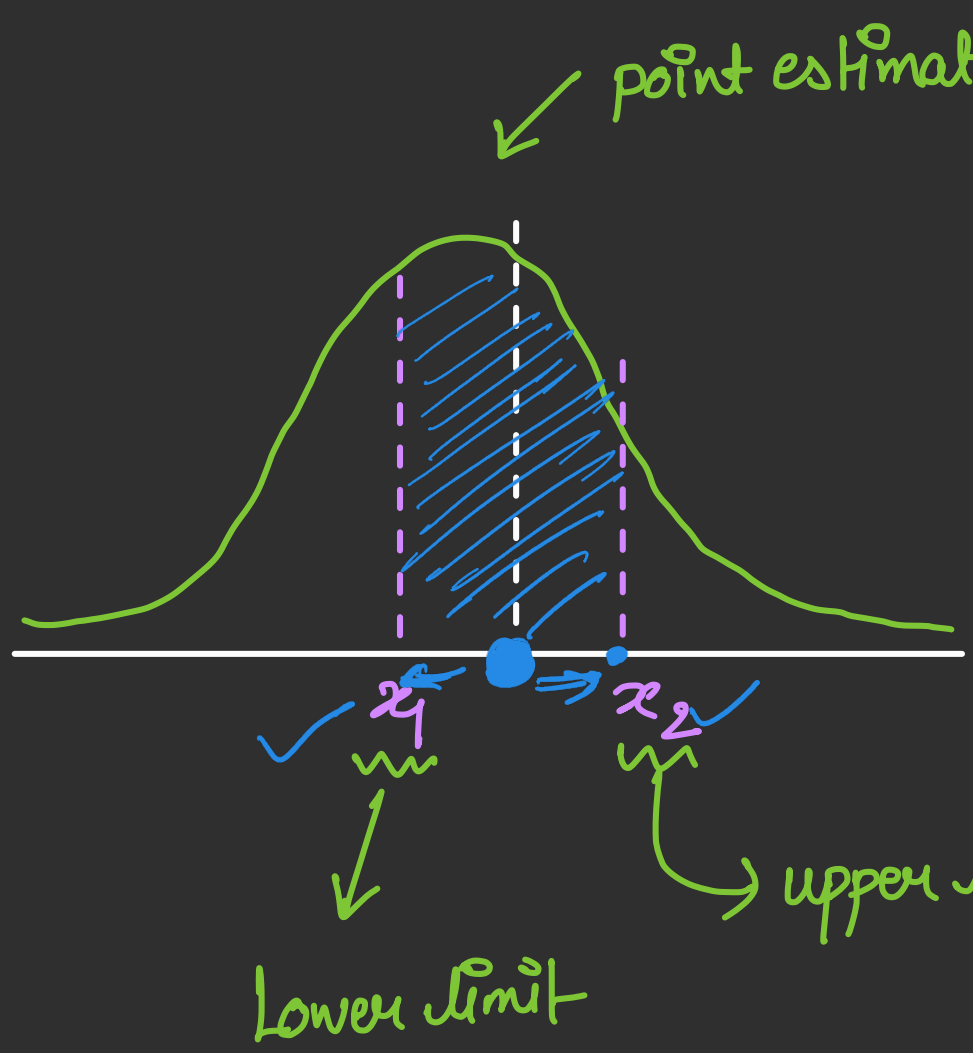
Confidence Interval



46.12 (point estimate)

Sample mean

How confident you are?



in 10 min

{ 11 am – 11:50 am

11 am – 12:00 am

$\bar{x}$ (means of sample mean)

$Z * SE$ → $(\sigma/\sqrt{n})$

CI = { point estimate $\pm$ Margin of error }

point estimate → mean value you got

Margin of error → ?

$ME = Z * SE$

$= Z \times (\sigma/\sqrt{n})$

C.I.

Q. How to calculate the confidence interval?

There are 2 ways to find confidence interval

Using CLT

This is for mean values. If we have mean as a statistic then we will calculate CI using the central limit theorem (CLT)

Using Bootstrapping

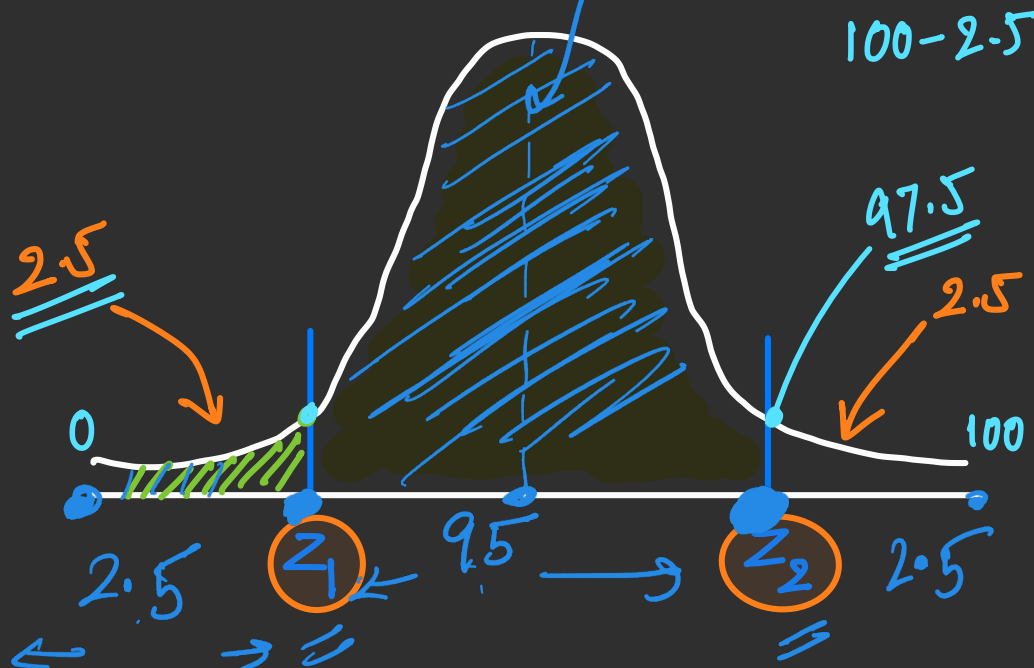
If you have statistics other than mean like median then we are not allowed to use CLT. We can use bootstrap method to calculate confidence intervals in those scenarios.

Confidence Interval Using CLT

100 - 95

5%

95%



95 out 100

95 + 2.5 = 97.5

100 - 95 = 5%

a) Calculate estimate with 95% CI?

100 - 2.5

97.5

norm.ppf(0.025)

z_1

norm.ppf(1 - 0.025)

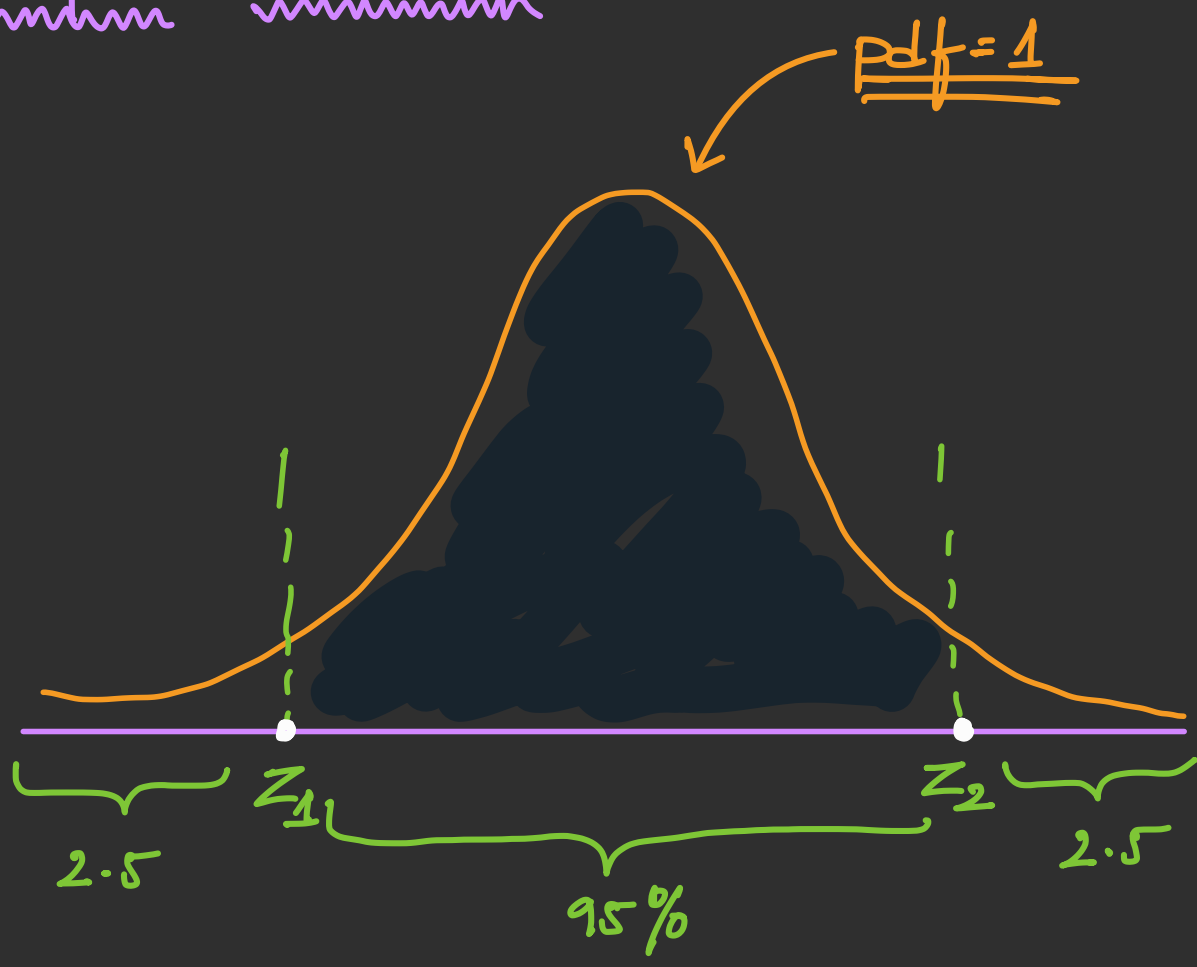
norm.ppf(97.5)

z_2

z_1 & z_2

$$CI = \text{popmean} \pm 1.95 * \frac{\sigma}{\sqrt{n}}$$

Compute 95% CI



$$\left[x = \mu + \left(Z_{samp} \times \frac{\sigma}{\sqrt{n}} \right) \right]$$

→ Margin of Error

$$CI = \underbrace{\text{point estimate}}_{\left(\begin{array}{l}\text{pop}^{n}\text{ mean} \\ \text{u} \\ \text{sample mean}\end{array}\right)} \pm \left\{ Z_{score} \times SE \right\}$$

$$CI = \text{point estimate} \pm \underbrace{\text{margin of error}}$$

$$CI = \mu \pm \left(z_{samp} \times \frac{\sigma}{\sqrt{n}} \right)$$

Confidence Interval using Bootstrap?

